How Does Chain-of-Thought Help Language Models Count?
A Causal Test of Global Aggregation and Sequential Retrieval

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Abstract

Counting sparse, semantically defined records in a long context requires a model to identify the relevant sources, avoid duplicates, aggregate their number, and express the result. We ask how explicit chain-of-thought (CoT) changes this computation relative to direct, non-thinking generation. Preliminary realistic needle-in-a-haystack experiments show that native CoT and controlled enumeration are substantially more accurate than direct answering in the moderate-count regime, while all modes degrade with count and context length. We formulate two competing accounts of direct counting—a record-by-record running update and a total formed near the final query—and contrast both with a sequential CoT computation consisting of matched-source retrieval, trace writing, progress update, and next-source or stop control. Our evidence standard separates count decodability, update consistency, and causal use. We then define functional scores for broad aggregation, targeted retrieval, and progress-transition components, and specify interventions that must propagate through source identity, unique coverage, progress-state quality, and the complete parsed answer. The mechanism study centers on Qwen3-8B and Gemma4-E4B, with a broader behavioral panel and controlled synthetic training dynamics.

TODO A0. Replace preliminary language with measured effect sizes and the strongest claim supported by the completed causal chain.

1 Introduction

Language models can retrieve individual facts from long contexts yet fail to report how many facts satisfy a semantic predicate. This sparse-record counting task is useful because an exact answer depends on several separable operations: semantic matching, coverage of all valid records, duplicate avoidance, accumulation, stopping, and numerical readout. A final error alone does not identify which operation failed.

Explicit reasoning often improves the answer, but the explanation is underspecified. CoT could improve source selection, externalize an otherwise internal state, supply convenient number tokens, alter the stopping policy, or simply add computation. The mechanistic question is therefore not whether "thinking helps," but:

> *Does CoT change the retrieval-and-aggregation computation used for counting, and which causal step accounts for the accuracy gain?*

Our working hypothesis is an *algorithmic path change*. Direct answering may combine many record contributions at the final query and read out a total from a terminal state. CoT may instead sequentialize the task into targeted retrieval, trace writing, progress update, and continue/stop decisions. Crucially, neither path is assumed in advance. In particular, we test whether direct answering actually maintains a causally used running counter, and whether a CoT progress state survives controls for visible index tokens.

This framing differs from homogeneous repeated-item counting and from systems that externally partition the input before aggregation. Here, the same unstructured context is reused across reasoning modes, and the relevant records are sparse semantic objects. The planned contributions are:

1. a frozen, request-level behavioral map over count, length, mode, model, and failure type;
2. a controlled sibling-stimulus design that separates record-wise running updates from terminal aggregation and removes visible-index leakage from CoT progress;
3. registered hidden-state landmarks and functional routing scores whose discovery and causal evaluation use disjoint prompt families; and
4. a time-indexed intervention chain linking source identity, unique coverage, progress and stopping, terminal readout, and the complete parsed answer.

2 Experimental Design

2.1 Task and frozen naturalistic stimuli

A passage contains source occurrences $R_1, \ldots, R_M$. Under a model tokenizer, occurrence i occupies the half-open token span $I_i = [s_i, e_i)$. A semantic predicate q gives

$$y_i = \psi_q(R_i) \in \{0, 1\},$$
$$G_q = \{i : y_i = 1\}, \qquad N = |G_q|.$$

The source identity is the *occurrence index* i, not merely the city or score string: retrieving the same occurrence twice is a trace duplicate, whereas two distinct valid occurrences both contribute to the count.

The behavioral suite is the frozen Realistic NIAH V2 grid. Filler text is built from a content-deduplicated 218-source Paul Graham corpus. For each seed, essays are shuffled once and the five length conditions use nested prefixes; an essay is never repeated to fill a context. We cross post-insertion canonical lengths $L \in \{2000, 3000, 5000, 10000, 20000\}$, counts $N \in \{1, 2, 3, 4, 5, 6, 8, 10, 20, 30\}$, and seeds $1234, \ldots, 1243$, yielding 500 base passages. Length is exact under the frozen Qwen3-8B tokenizer; realized length under every evaluated tokenizer is also recorded.

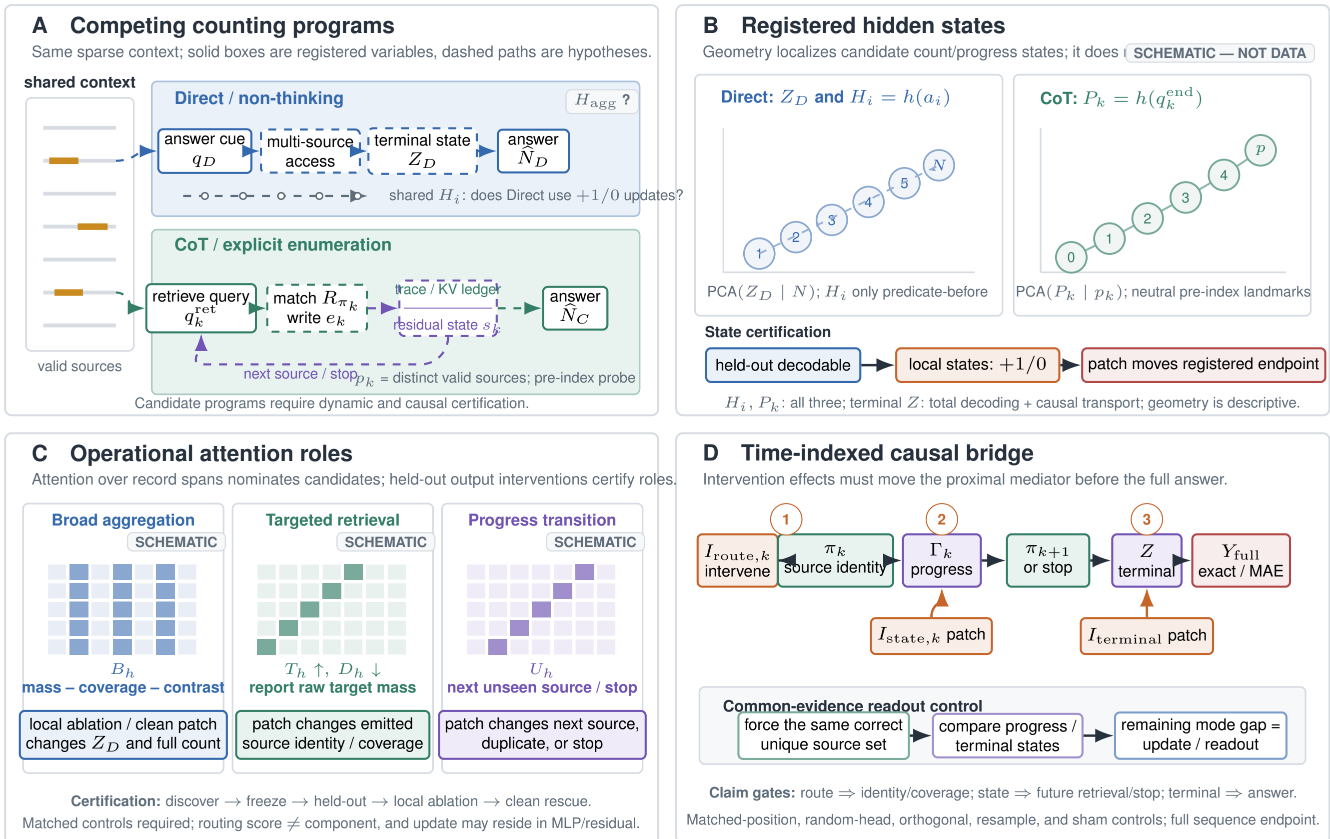

Figure 1: Mechanism map and causal test plan for Direct and CoT counting. **A:** Direct answering is tested for terminal multi-source aggregation and, only in prompts where the counting predicate precedes the records, whether it causally uses the shared record-end states H_i. CoT is tested for a recurrent program that retrieves one matched record, writes evidence, updates a progress carrier, and selects the next record or stops. **B:** Registered hidden-state landmarks separate the Direct terminal state Z_D, predicate-conditioned record-end states H_i, and CoT pre-index states P_k. Local counter/progress claims require held-out decoding, the correct $+1/0$ dynamics, and a downstream patching effect; terminal states require total decoding and causal transport. **C:** Broad-aggregation, targeted-retrieval, and progress-transition candidates are nominated by attention scores but named functionally only after local ablation and clean patch rescue. **D:** Time-indexed interventions must first move the corresponding mediator—source identity/coverage, future retrieval or stopping, or terminal readout—and then the complete parsed answer. Common-evidence controls isolate update/readout from retrieval. All miniature trajectories and heat maps are schematic analysis templates, not empirical results; dashed paths are hypotheses.

Each target is a sentence of the form

```
In the 2024 city score audit, <city>
received a score of <score>.
```

Cities and scores are unique within a passage. Candidate insertion points are conservative sentence endings that exclude numeric strings, URLs, paths, and common abbreviation periods. They are sampled without replacement when enough sites exist; any replacement or word-boundary fallback is logged rather than silently treated as an ordinary sentence insertion. The generator then adjusts only neutral filler and regenerates until the *post-insertion* canonical length is exact. Freezing fails unless every target appears exactly once, no target-like sentence occurs in the filler, every character span maps to a verified tokenizer span, and no insertion is truncated. The base passage, target identities, insertion depths, and SHA256 are frozen before mode or model expansion. Consequently, all modes and models for one stimulus see the identical passage text.

2.2 Controlled sibling stimuli for mechanism identification

The naturalistic grid measures behavior but is not by itself sufficient to distinguish a local counter from late aggregation: changing N also changes how many record-like sentences are inserted. We therefore construct a disjoint, model-audited mechanism suite. A stimulus family

$$\omega = (H, q, \{u_j, I_j, y_j\}_{j=1}^{K})$$

contains the same filler H, $K = 10$ fixed record-like slots, opaque occurrence IDs u_j, and positive/negative variants that differ only in one predicate-critical attribute (balanced across wrong-audit-type, e.g. `city safety audit`, and wrong-document-type, e.g. `city score report`, hard negatives; year alone is never treated as negative because the frozen predicate does not restrict year). Within each mechanism model, rejection sampling makes every positive/negative pair equal in token length. Slots are sampled once from ten normalized-depth strata at legal sentence boundaries, with a target minimum separation of 128 model tokens; their text, order, and positions remain fixed across all siblings at one length. A balanced permutation assigns which N slots are positive, so neither depth, hard- negative type, nor a particular city predicts validity. Longer siblings change only neutral filler and preserve slot order and depth strata.

Mechanism corpora are rendered separately for the Qwen and Gemma tokenizers so the post-insertion passage length is exact and every within-model sibling retains identical downstream token positions. Cross-model corpora share the semantic family manifest, slot ordering, labels, and normalized depths, but are not claimed to have identical token IDs or absolute token indices.

Two paired manipulations identify different computations. A *prefix-shift* sibling swaps an early positive and a late matched negative, preserving final N, L, and all slot identities while changing the cumulative count between those slots. A *total-shift* sibling flips one fixed slot from negative to positive, producing $N \leftrightarrow N + 1$ without changing sequence length or location. Clean/corrupt activation-patching pairs are instances of these siblings; the complete rendered token sequence is equal length within a model, and every modified span is stored in the manifest.

TODO D1. Freeze the controlled mechanism suite at $N \in \{0, \ldots, 6, 8, 10\}$. At the primary length $L = 5$k, target at least 120 independent families assigned before capture to 60 discovery, 20 validation, and 40 locked test families; use a family-cluster pilot power analysis to increase, not silently shrink, the locked set. After all layers, heads, and thresholds are frozen, test length transfer on at least 40 new locked families at each of $L = 2$k and 10k. Store positive/negative siblings, prefix-shift and total-shift pairs, prompt and continuation token IDs, every model-specific character/token span, and contamination and length-alignment audits. Prefix-shift is eligible only for $1 \leq N \leq K - 1$; at $N = 0$ or $N = K$, use the available one-direction total shift rather than inventing a swap. Use seeds disjoint from the behavioral grid.

2.3 Paired prompts, reasoning policy, and decoding

The registered behavioral layout is *predicate-before*: the exact common cue "You will need to count all city-score audit records . . . " and its definition precede the passage, while the response request follows it. Thus record-end states in the existing V2 data have already seen the predicate even though the final question is after the passage. The passage and question sentence are reused verbatim; the remaining response-policy text and the model's frozen reasoning switch define the mode.

In the frozen V2 layout, the mode-specific response suffix occurs *after* the passage. Under a causal mask, passage-token states therefore cannot differ across modes for the same checkpoint and stimulus if their rendered prefixes through the passage are identical. We audit this prefix identity after applying each model's chat template. Such states are treated as a shared scanning substrate, not as evidence that CoT encoded a cleaner counter while reading. The mode comparison instead registers the Direct terminal state after its suffix and the CoT states created during trace generation. A separate *policy-before* control moves the Direct/enumeration instruction before the passage to test whether an anticipated trace can change prompt encoding; it is never pooled with the original behavioral grid.

The mechanism study adds two controls. First, a 2×2 prompt-policy factorial crosses visible instruction (Direct versus unindexed enumeration) with the Qwen/Gemma reasoning switch (off versus on), separating trace format from the native thinking policy. Second, predicate placement has three levels: the registered predicate-before layout, a *predicate-after* layout that moves the entire task definition after the passage, and a *predicate-repeat* layout that repeats it near generation. Predicate-after record states cannot encode the task-specific cumulative count and form a causal-mask negative control; predicate-repeat separates availability during encoding from instruction recency.

Activations on a fixed prefix are deterministic. For free-continuation causal endpoints, paired clean, corrupt, and patched runs use the same generation seed (common random numbers) and retain all parse failures and truncations. Data-generation seeds and decoding seeds are separate fields; the ten behavioral seeds are stimulus replicates, not estimates of sampling variance. We report both teacher-forced log likelihood of the complete registered continuation and free-generation exactness/MAE; a first-digit logit is never the sole endpoint. Equal-length null scratchpads control for added tokens and recurrent computation.

TODO D2. Run the Qwen3-8B/Gemma4-E4B 2×2 instruction-by-policy control and the three predicate-placement conditions on the locked mechanism families, plus policy-before versus suffix-after placement. Use the same sampler, generation seed, and maximum budget within each reasoning-toggle contrast. Freeze chat-template revisions and stop rules; hash rendered prefixes through the final passage token. Use at least three paired decoding seeds for stochastic free-continuation endpoints while retaining teacher-forced sequence likelihood as the variance-free endpoint. Verify that any claimed algorithmic change is not explained only by instruction recency, output length, or a checkpoint that reasons in every nominal mode.

2.4 Analysis unit, splitting, and averaging

The atomic representation observation is *one stimulus family* $\times$ *one rendered prompt variant* $\times$ *one registered landmark* $\times$ *one layer*; the independent unit for splitting and uncertainty is the stimulus family. We do not replace these observations by cross-prompt mean states before fitting a probe or PCA, or before a primary activation patch. All siblings—modes, predicate placements, N/L variants, and clean/corrupt counterfactuals—are assigned to the same split by family ID. Discovery fits preprocessing and selects layers/heads; validation freezes hyperparameters and thresholds; the test families are opened once for the registered comparisons. Additional evaluation holds

Mode	Response-policy suffix	Switchable-model policy	Primary role
DIRECT	One line, `Total: <integer>`; explicitly forbids explanation or listing	thinking off; greedy; 64 tokens	non-thinking endpoint
ENUM	Passage-order bullet list of actual `city: score` pairs, then total	thinking off; greedy; 1536 tokens	aligned, index-free process control
INDEX-ENUM	Same list with ordinary numeric indices, then total	thinking off; greedy; 1536 tokens	visible-index control
NATIVE-COT	Concise native reasoning, stop once the count is known, then the same total line	thinking on; frozen model-specific sampling; 4096 tokens	ecological CoT condition

Table 1: Prompt control. The common cue, passage, question, chat-template revision, and parser are frozen within a checkpoint. Always-on reasoning checkpoints are not interpreted as non-thinking in their nominal Direct cell.

out record lexemes, templates, and complete (N, L) cells.

Let $\mathcal{B}$ be the stimulus families, $\mathcal{V}_b$ the eligible rendered prompt variants of family b, and J_{bv} the eligible landmarks in variant v. Representation losses use

$$\frac{1}{|\mathcal{B}|}\sum_{b\in\mathcal{B}}\frac{1}{|\mathcal{V}_b|}\sum_{v\in\mathcal{V}_b}\frac{1}{|J_{bv}|}\sum_{j\in J_{bv}}\mathcal{L}\big(g(h^{\ell}_{bvj}), y_{bvj}\big)\,.$$

Thus every family has unit total weight, every variant within a family has equal weight, and multiple record or trace landmarks sum to one within a variant. PCA centering, nuisance residualization, decoder regularization, and count directions are fitted on discovery prompts only. Figures may show held-out individual points and class centroids with family-cluster bootstrap intervals, but centroids are summaries, not substituted observations. Natural-state patching uses one matched donor prompt; a training-set mean intervention is reported only as an ablation control. Steering is the only causal intervention whose direction is defined by a cross-prompt mean, and that mean is estimated from paired discovery-set differences. Attention statistics are first averaged over eligible steps within a variant, then over variants within a family, and finally equally over families. We do not average raw hidden vectors across layers, checkpoints, token roles, or model families.

3 Behavioral Phenomenon

3.1 Outcomes and paired estimands

Direct versus Native-CoT is the primary behavioral *interface contrast*: in the frozen suite it changes both the visible instruction and the model's reasoning policy, so it is not interpreted as a single-variable causal effect of CoT. The two enumeration modes separate the benefit of an explicit retrieval ledger from unrestricted native reasoning. The unit of pairing is the frozen 500-stimulus ID. Primary outcomes are intent-to-treat exact-count accuracy, signed error, absolute error, parse success, truncation, and protocol compliance. *Exact-count accuracy* means that the last valid parsed `Total: <integer>` equals N; *registered success* additionally requires the mode-specific output format and no length truncation. We report both and never call registered success "accuracy" without qualification. For a trace, let $\widehat{G}$ be the emitted multiset of aligned source-occurrence IDs and let $A = \widehat{N} - |\widehat{G}|$. Then

$$\widehat{N} - N = \underbrace{\mathrm{FP} - \mathrm{FN} + \mathrm{Dup}}_{\text{retrieval/coverage}} + \underbrace{A}_{\text{aggregation/protocol}}, \qquad (1)$$

where $\mathrm{Dup} = |\widehat{G}| - |\operatorname{uniq}(\widehat{G})|$. This identity is applied to aligned enumerations, with $\mathrm{FP} = |\operatorname{uniq}(\widehat{G}) \setminus G_q|$ and $\mathrm{FN} = |G_q \setminus \operatorname{uniq}(\widehat{G})|$. It is evaluated only when both the total and source identities are parseable; every other trace remains an intent-to-treat parse, alignment, or protocol failure rather than being dropped. Mode contrasts use paired stimulus effects and stimulus-family-cluster bootstrap intervals.

3.2 Behavior and mechanism panels

In the current core slice ($N \le 10$, $L \le 10$k), the working summary reports means of 0.580 for DIRECT, 0.962 for ENUM, 0.971 for INDEX-ENUM, and 0.975 for NATIVE-COT across four checkpoints. These values remain explicitly preliminary until B0 verifies the metric label and source version for every row. Individual checkpoints are heterogeneous: the present evidence supports a broad advantage for trace-generating interfaces, not a claim that Native-CoT is best for every model. Current results also suggest lower accuracy and larger absolute error as either N or L grows, but the interaction and coefficient stability have not yet been established.

TODO B0. Audit every requested checkpoint against the frozen stimulus hash, tokenizer and chat-template revision, reasoning policy, generation budget, parser version, request-level failure flags, and exact-count versus registered-success field. Preserve the per-row `source_version`: the current report combines formal-v2 Direct/Native rows with v2.1 enumeration replacements. Treat GLM-4-9B and GLM-Z1-9B as distinct checkpoints rather than a same-weight thinking toggle, and record the precise reason any model/mode is not mechanism-eligible.

TODO B1. Complete the requested behavior model set, including the missing Qwen 14B slot, on the same 500 frozen stimuli and four registered modes. Report paired effect sizes over the full grid and the moderate regime; intent-to-treat exact accuracy, conditional numerical accuracy, MAE, signed

Panel	Checkpoints	Scope
Behavior / optional law	Qwen3-4B, 8B, 14B, and 32B; Gemma4-E4B; GLM-4-9B and GLM-Z1-9B; Cogito-v1-8B; Nemotron-Nano-v2-9B; Nemotron-3-Nano-4B	Mode effects, degradation with N and L, family heterogeneity, and complete failure budgets. Qwen within-family trends are primary; pooled cross-family size trends are secondary.
Mechanism	Qwen3-8B and Gemma4-E4B	Registered residual states and head outputs; steering; state, head-output, and path patching; query-local ablation; semantic outcome analysis.
Synthetic dynamics	Small causal transformers in one frozen v20-based setting with a v15-style all-sequence control	When decodability, valid updates, causal use, and routing roles emerge during training.

Table 2: Experimental organization. Mechanism roles are aligned by semantic function, not by layer or head index. Models that cannot implement a registered reasoning interface remain in the failure appendix rather than being silently removed.

error, parse/truncation/protocol failures, and Equation (1). Use stimulus-cluster bootstrap intervals and do not add predicate- or policy-placement variants to this already frozen behavior grid.

3.3 Response surfaces are secondary

We do not force a universal scaling law across dense and mixture-of-experts models with different tokenizers and post-training. A candidate family-specific response surface is

$$\text{logit } \Pr(\widehat{N} = N) = \alpha_{f,m} + s_{f,m}(\log L, \log(N+1)),$$

where f and m denote family and mode. Linear, log-linear, interaction, and shape-constrained candidates are compared on held-out stimulus families.

TODO B2. Pre-register candidate surfaces and grouped train/validation/test splits. Promote a quantitative law only if it improves held-out log loss and Brier score over a condition-only baseline and remains stable across seeds and templates. Otherwise report qualitative partial trends with uncertainty and move coefficient tables to the appendix.

4 Competing Computations

4.1 Minimal computation graphs

We use minimal equations to separate what is being tested from how a particular transformer implements it. For direct answering,

$$\begin{aligned} w_i^D &= \mathcal{R}_D(q, R_i), \\ Z_D &= \mathcal{G}_D\left(\{w_i^D V_i\}_{i=1}^M\right), \\ \widehat{N}_D &= \mathcal{Q}_D(Z_D). \end{aligned} \tag{2}$$

$\mathcal{R}_D$ scores or routes record information, $\mathcal{G}_D$ constructs a terminal task state, and $\mathcal{Q}_D$ reads out the count. Equation (2) does not assume that $\mathcal{G}_D$ is a sum, a density code, a running counter, or a single attention head. It supports two competing hypotheses:

H_{run}**:**
: a query-conditioned state updates by approximately $+1$ after a valid source occurrence and by 0 after a matched non-target, and this state affects later computation;

H_{agg}**:**
: local states may contain content or ordinal information, but the task-relevant total is constructed mainly near the final query through multi-record access and terminal readout.

For CoT or explicit enumeration,

$$\begin{aligned} \pi_k &= \Pi_C(q, \Gamma_{k-1}), & e_k &= \mathcal{W}_C(R_{\pi_k}), \\ \Gamma_k &= \mathcal{F}_C(\Gamma_{k-1}, e_k), & \widehat{N}_C &= \mathcal{Q}_C(\Gamma_K). \end{aligned} \tag{3}$$

π_k is the source selected at step k, e_k is the written item, and Γ_k is the progress carrier used to choose another source or stop. Γ_k may combine a residual-stream state s_k with an external ledger in the visible trace and KV cache; it is not assumed to be a scalar neural counter.

After source alignment, let $\pi_k \in \{1, \dots, M\} \cup \{\perp\}$ be the source-occurrence identity at trace step k, with $\perp$ denoting an unaligned item. The visited set and semantic progress label are

$$\mathcal{V}_k = \text{uniq}\{\pi_u : 1 \le u \le k,\ \pi_u \ne \perp\}, \qquad p_k = |\mathcal{V}_k \cap G_q|, \tag{4}$$

the number of distinct valid sources actually retrieved. It is deliberately different from the visible step number k.

4.2 Registered token landmarks and activation capture

Let h_t^ℓ be the post-block residual stream at token t: the output of transformer block ℓ after its residual addition and before normalization by the next block. Embeddings ($\ell = 0$) are diagnostic only. Every landmark is located by character-to-token offset mapping on the *final model-specific chat-rendered sequence*; no position is inferred by adding a standalone needle length. The mechanism generator appends the same invariant end-of-record string D_{eor} = `[END-RECORD]` followed by a newline to every positive and matched-negative block. The record-end landmark is the last token whose offset overlaps the closing bracket, and a pair is rejected unless that landmark has the same token index in both siblings. The naturalistic V2 sensitivity analysis instead uses the final token overlapping the target sentence's terminal punctuation.

The chat prompt is tokenized with the exact generation-time template. A teacher-forced continuation is tokenized with its generation-time leading whitespace and no additional special tokens, then appended as continuation token IDs; we never re-tokenize a raw prompt–continuation string in a way that can change the final prompt token. Continuation offsets are computed on that token axis and shifted by the frozen prompt length. A sample is rejected if the token overlapping the `Item:` colon also overlaps the first source character, or if the token overlapping the `Total:` colon also contains a count character. The manifest saves the complete prompt and continuation token IDs.

Writing a_i for the registered record-end token, q_k^{ret} for the `Item:` colon position, and q_k^{end} for the corresponding D_{eor} landmark, the state notation is $H_i^\ell = h_{a_i}^\ell$, $Q_k^\ell = h_{q_k^{\text{ret}}}^\ell$, and $P_k^\ell = h_{q_k^{\text{end}}}^\ell$. Likewise, q_D is the colon position in the Direct `Total:` cue and $Z_D^\ell = h_{q_D}^\ell$.

Controlled CoT prefixes use the literal grammar `Item: <city>: <score> [END-RECORD]`. The unindexed grammar is primary; numeric, random nonnumeric, and permuted labels are placed outside the registered `Item:` and D_{eor} tokens. We teacher-force all prefixes needed to create the same k with different p_k, then let the model continue freely for causal endpoints. Native free-running traces are aligned to source-occurrence spans post hoc and serve as ecological replication, not landmark discovery; parse failures remain failures.

Predicate availability, rather than the location of the final question alone, determines the local-counter estimand. In predicate-before and predicate-repeat prompts,

$$c_q(t) = \sum_{i:e_i \le t} \psi_q(R_i)$$

is available at record endpoints. In predicate-after prompts it is forbidden by the causal mask, so H_i^ℓ is used only as a query-independent content control. Terminal Z_D^ℓ and final-query source access are tested in all three layouts.

4.3 When we use the word counter

A candidate state is called a *counter* only after three nested tests:

1. **Decodable:** a decoder trained on disjoint prompts predicts $c_q(t)$, p_k, or N from the registered state.
2. **Dynamically consistent:** Direct decoded state changes by $+1$ after a valid input occurrence and 0 after a matched non-target; CoT progress changes by $+1$ after a newly retrieved valid source and 0 after an invalid or repeated source. Both tests control position, length, total count, density, and identity.
3. **Causally used:** a natural-state patch or controlled steering intervention changes downstream source choice, progress, stopping, or the complete final answer in the predicted direction.

PCA is a visualization of candidate geometry; it does not satisfy any of these tests by itself.

5 Registered Count States and Causal Control

5.1 Decoding and geometry protocol

For each layer and registered role, we fit a ridge decoder on individual discovery prompt-landmark states with the family-equal weights defined above. Regularization is selected on validation families. A nuisance-only model $g_0(u)$ is compared with a state-augmented model $g_1(h^\ell, u)$; the evidence attributed to the representation is held-out improvement in MAE and R^2, not raw predictability of a count that is already correlated with position. For Direct local states,

$$u_D = (i, K, N, L, a_i/L, N/L, \text{depth bin}, \text{template ID}),$$

while for CoT progress,

$$u_C = (k, N, L, \text{trace-token length}, \text{visible-label identity}).$$

Remaining count $N - c_i$ is reported as a separate outcome rather than placed in the same nuisance vector as N and c_i.

For visualization, one shared PCA basis per model/layer is fitted after balancing the discovery sample over registered role, mode, semantic label, and (N, L) cell and recording the downsampling seed. The frozen basis projects individual locked-test states; mode-specific PCA is a sensitivity analysis. Correct and incorrect outputs are both included in the primary fit and are separated only for a pre-registered diagnostic. Thus a clean-looking trajectory cannot be created by averaging states within a prompt or by selecting successful traces.

TODO R1. At every landmark in Table 3, fit family-weighted nuisance-only and state-augmented decoders for Qwen3-8B and Gemma4-E4B. Freeze preprocessing, ridge penalties, PCA bases, and layer selection before the locked test. Report held-out incremental R^2/MAE, unseen identity/template and (N, L) transfer, state dispersion, and correct/error strata. Plot individual test points plus family-cluster centroids; do not fit a projection to the presentation prompts.

5.2 Direct: running update or terminal aggregation?

For predicate-before and predicate-repeat prompts, we extract H_i^ℓ at all K controlled slots, not only the positive records. The primary dynamic test is a same-position prefix-shift contrast, not the difference between two positions inside one prompt. Let siblings A and B exchange a positive slot at r with a matched negative at $s > r$, and let g_h^ℓ be a hidden-only decoder applied after discovery-set nuisance residualization. At the same landmark j,

$$\delta_j^\ell = g_h^\ell(\widetilde{H}_j^{A,\ell}) - g_h^\ell(\widetilde{H}_j^{B,\ell}) \overset{H_{\text{run}}}{\simeq} \begin{cases} 0, & j < r, \\ 1, & r \le j < s, \\ 0, & j \ge s. \end{cases}$$

The confirmatory region uses unchanged record blocks strictly between r and s, so the local token and record identity are

Registered variable	Exact primary token	Label and causal role
Direct local H_i^ℓ	Token overlapping the closing bracket of $D_{\rm eor}$ after slot i; predicate-before and predicate-repeat only	$c_i = \sum_{j\le i} y_j$; test target $+1$, matched non-target 0, and downstream propagation
Direct terminal Z_D^ℓ	Token overlapping the colon in a teacher-forced assistant prefix `Total:`, immediately before the first count token	Total N; terminal aggregation and complete-answer readout
CoT retrieval Q_k^ℓ	Token overlapping the colon in a fixed, unindexed `Item:` marker immediately before the source-bearing city at step k	Gold next source identity g_k; targeted retrieval
CoT progress P_k^ℓ	Token overlapping the closing bracket of $D_{\rm eor}$ after the kth source mention and before any next index or label	Semantic progress p_k; next-source and continue/stop control
Controlled CoT terminal $Z_C^{\mathrm{ctrl},\ell}$	Token overlapping the colon in `Total:` after the canonical teacher-forced trace	Primary terminal readout given registered evidence
Native CoT terminal $Z_C^{\mathrm{nat},\ell}$	Parsed `Total:` colon in a free-running native trace, when uniquely aligned	Ecological replication only; never pooled with $Z_C^{\mathrm{ctrl},\ell}$

Table 3: Pre-registered hidden-state landmarks. Primary analyses use one token per landmark, not a record-span average. Record-span sums are used for attention, where the estimand is source access rather than a local state.

identical while the correct prefix count differs by one. The endpoints at r and s and adjacent-position increments within one prompt are secondary content/update diagnostics. This contrast fixes final N, record-like block count, length, and position and is therefore the decisive guard against total-count, density, and nuisance-only leakage. Predicate-after states provide the causal-mask negative control.

Under the frozen suffix-after V2 layout, a successful prefix-hash audit makes each H_i^ℓ identical across response modes for a given checkpoint and stimulus. Hence this analysis asks whether DIRECT later *uses a shared scanning state*; it cannot establish that DIRECT created a mode-specific state or that CoT created a cleaner one while reading. Mode-specific passage encoding is tested only in the separate policy-before control.

For causal propagation, a total-shift clean/corrupt pair differs at one equal-length slot r. The primary counter-transport patch copies a clean natural activation at the first unchanged downstream record landmark $j > r$ into the corrupt run at the same token and continues the forward pass. We pre-register flips with such a downstream anchor; patching the modified record's own endpoint is retained only as an evidence-transport positive control. Endpoints are, in order: (i) decoded cumulative count at later slots; (ii) the terminal state Z_D; (iii) sequence log likelihood after `Total:`; and (iv) the complete free-generated count. A positive-to- positive identity patch, negative-to-negative patch, unrelated-position patch, norm-matched random state, and clean-to-clean sham estimate non-specific transport.

$H_{\rm agg}$ requires positive evidence and is not accepted merely because a local probe fails. We test whether final `Total:` states and query-local multi-record routes carry N, whether total-shift terminal patches move the full answer, and whether these effects remain when local ordinal information is controlled. Evidence for $H_{\rm run}$ requires decodability, correct $+1/0$ dynamics, and propagated local patches; evidence for $H_{\rm agg}$ requires a selective terminal route/state intervention.

TODO R2. Run the Direct prefix-shift and total-shift experiments at every frozen candidate layer. Use same-position hidden-only contrasts at unchanged landmarks for the primary $+1/0$ test; do not treat a second valid input occurrence as a no-op. Patch single natural donors at unchanged downstream H_j and at Z_D separately, with the modified-record endpoint only as an evidence-transport control. Measure later landmarks, terminal state, complete-answer likelihood, and free-generation exact/MAE. Include predicate-after and predicate-repeat, identity-preserving, position, random, norm, and sham controls.

5.3 CoT progress without label leakage

The CoT label is p_k from Equation (4), not the emitted step number. In passage-order enumeration, the pre-registered next gold source is

$$g_k = \operatorname*{arg\,min}_{i\in G_q\setminus\mathcal{V}_{k-1}} s_i,$$

where $\mathcal{V}_{k-1}$ is the set of source-occurrence IDs already retrieved. Gold-target alignment is the primary estimand; attention to the source the model actually emits is reported separately as execution alignment, so a consistently wrong retrieval does not receive a high mechanism score. If the set is empty, the registered target is `stop` rather than a source.

At the same visible step k, we teacher-force three equal-length prefixes whose newest item is: a new gold source ($\Delta p = 1$), a matched non-target ($\Delta p = 0$), or a repetition of an already visited source occurrence ($\Delta p = 0$). Complementary prefixes hold p_k fixed while changing trace extent. We repeat them with no index, random nonnumeric labels, and permuted numeric indices, always extracting at the invariant `Item:` and $D_{\rm eor}$ tokens. A state that follows p_k under these transformations is evidence for semantic progress; a state that follows k or the displayed label is not.

A clean P_k^ℓ is patched into a matched corrupt prefix before the model chooses its next source. The proximal endpoints are the sequence probability and free-run identity of g_{k+1}, duplicate probability, and continue/stop probability; later endpoints are unique coverage, P_{k+1}, and the complete count. Residual-only transport patches P_k^ℓ while preserving the recipient trace/KV cache. Trace/KV-only path patching transports donor trace-token keys and values while preserving the recipient current residual. The joint intervention tests whether the two carriers are complementary; it is interpreted cautiously because KV transport is the more off-manifold operation.

For geometric steering, the discovery-set direction is the normalized prompt-paired difference

$$v_\ell = \frac{\mathbb{E}_{\omega,k}[P_k^\ell(\Delta p = 1) - P_k^\ell(\Delta p = 0)]}{\left\|\mathbb{E}_{\omega,k}[P_k^\ell(\Delta p = 1) - P_k^\ell(\Delta p = 0)]\right\|_2}.$$

We add αv_ℓ only at the registered token, choose the layer and α range on validation families, and compare with norm-matched random and orthogonal directions. This cross-prompt mean defines a steering direction only; primary patching still uses a single matched natural donor.

TODO R3. Run the equal-length new-gold/non-target/repeated-source prefix experiment with unindexed, random-label, and permuted-index interfaces. Test same-k/different-p_k and same-p_k/different-extent pairs; residual-only, trace/KV-only, and joint transport; natural same-position patches; and a validation-selected α steering sweep. Report gold next-source likelihood/identity, duplicate and stop behavior, later progress, unique coverage, and the complete parsed count with norm-matched, random, orthogonal, position, and sham controls.

5.4 State dispersion is not probe error

Let $\widetilde{h}^\ell$ be the residual after a discovery-set nuisance model removes the registered position, length, density, template, and label covariates. For semantic count or progress c, define the family-equally weighted moments for each registered role separately

$$\mu_c^\ell = \mathbb{E}_\omega[\widetilde{h}^\ell \mid c], \qquad \Sigma_c^\ell = \mathrm{Cov}_\omega(\widetilde{h}^\ell \mid c).$$

Let $d_c = \mu_{c+1}^\ell - \mu_c^\ell$, $u_c = d_c/(\|d_c\|_2 + \varepsilon)$, and $\bar{\Sigma}_c = (\Sigma_c^\ell + \Sigma_{c+1}^\ell)/2$. We measure conditional state dispersion relative to adjacent separation as

$$\nu_{\text{state}}^\ell(c) = \frac{u_c^\top \bar{\Sigma}_c u_c}{\|d_c\|_2^2 + \varepsilon}. \tag{5}$$

This is a representation statistic and is reported only when adjacent separation exceeds a validation-frozen threshold. Held-out decoder error measures accessibility to a chosen decoder and is reported separately. We test, rather than assume, that $\nu_{\text{state}}(c)$ grows with count in Direct and is smaller at CoT progress landmarks. Because H_i, P_k, and Z inhabit different token roles, this cross-role comparison is exploratory and cannot by itself establish an algorithmic path change.

6 Routing and Update Components

6.1 Three functional scores

Let $A_{q,t}^{\ell,h}$ be attention from query position q to token t for head h, and define record-span mass

$$a_i^{\ell,h}(q) = \sum_{t \in I_i} A_{q,t}^{\ell,h}.$$

We sum over the complete, pre-registered record span rather than selecting a favorable token after inspecting a heatmap. Each score is computed per eligible query, averaged over queries within a prompt variant, over variants within a family, and only then equally over families. The following scores nominate candidates; causal criteria below determine the functional label. Because raw mass can favor a longer city or record span, every score is accompanied by the span-density sensitivity $a_i^{\ell,h}(q)/|I_i|$ and enrichment relative to a uniform-attention baseline.

Broad aggregation score. At the Direct answer-cue token q_D, let $M_G^{\ell,h} = \sum_{i \in G_q} a_i^{\ell,h}(q_D)$ and $r_i^{\ell,h} = a_i^{\ell,h}(q_D)/(M_G^{\ell,h} + \varepsilon)$. For $N \geq 2$, define

$$B_{\ell,h} = \mathbb{E}_\omega\left[M_G^{\ell,h}\,\frac{\exp\{-\sum_{i \in G_q} r_i^{\ell,h}\log(r_i^{\ell,h} + \varepsilon)\}}{N}\right].$$

The effective-coverage factor is one when gold-span mass is uniform and $1/N$ when it is concentrated on one source. We additionally report gold-versus-matched-negative contrast; high $B_{\ell,h}$ without contrast is nonspecific broad access. We call a high-scoring head a *broad aggregation component* only if query-local head-output ablation disrupts Z_D or the count and a clean output patch restores it.

Targeted retrieval score. At CoT retrieval step k, let g_k be the pre-registered next gold source in passage order and $\mathcal{V}_{k-1}$ the sources already present in the prefix. Define

$$T_{\ell,h} = \mathbb{E}_\omega \mathbb{E}_{k|\omega}\left[\frac{a_{g_k}^{\ell,h}(q_k^{\text{ret}})}{\sum_{i=1}^K a_i^{\ell,h}(q_k^{\text{ret}}) + \varepsilon}\right],$$

$$D_{\ell,h} = \mathbb{E}_\omega \mathbb{E}_{k|\omega}\left[\frac{\sum_{i \in \mathcal{V}_{k-1}} a_i^{\ell,h}(q_k^{\text{ret}})}{\sum_{i=1}^K a_i^{\ell,h}(q_k^{\text{ret}}) + \varepsilon}\right].$$

We report raw gold-target mass alongside conditional selectivity. Attention to the source the model actually emits is a separate execution-alignment statistic. A targeted retrieval component must have high gold-target selectivity, low visited-source mass, and a causal effect on the emitted source identity.

Progress-transition score. For nonterminal controlled prefixes, define

$$U_{\ell,h} = \mathbb{E}_\omega \mathbb{E}_{k|\omega}\left[\frac{a_{g_{k+1}}^{\ell,h}(q_k^{\text{end}})}{\sum_{i=1}^K a_i^{\ell,h}(q_k^{\text{end}}) + \varepsilon}\right].$$

$U_{\ell,h}$ screens for a component that uses the completed step to prepare the next gold source. Terminal prefixes are evaluated by the probability of `Total:`/stop and by free-running stopping behavior rather than forcing a nonexistent source target. We use *progress-transition* or *next-source control*, not "successor head," unless relabeling shows a token-independent $k \mapsto k+1$ OV map (Gould et al., 2023). The update itself may be implemented by an MLP or residual stream even when an attention head supplies routing.

6.2 From heatmap to functional component

Every role follows the same validation ladder:

$$\text{discover} \rightarrow \text{freeze} \rightarrow \text{held-out selectivity} \\ \rightarrow \text{local ablation} \rightarrow \text{clean patch rescue.}$$

Attention maps are retained as interpretable summaries, but attention alone does not establish information flow (Zhang and Nanda, 2024). Ablations are applied to head outputs at registered queries—the head contribution after its value aggregation and output projection—not globally whenever a local intervention is possible. Controls include random and same-layer heads, attention-mass-matched heads, lexical adjacency, resample and mean ablation, an unrelated-token local ablation, and clean-to-clean/corrupt-to-corrupt sham patches.

TODO H1. On discovery families, compute $B_{\ell,h}$, $T_{\ell,h}$, $D_{\ell,h}$, and $U_{\ell,h}$ for every layer/head in both mechanism models using the step–variant–family aggregation above. Freeze candidates and thresholds on validation families before the locked test; pre-register at most five candidates per role and model and use a family-wise max-statistic permutation screen. Separate gold-target from emitted-target alignment; plot raw span mass, contrast/selectivity, and a small number of real test heatmaps. Apply Holm correction across the frozen causal head/endpoints, not across post-selected heatmaps.

TODO H2. For each frozen candidate, run query-local head-output ablation, clean-to-corrupt patching, and path patching into the registered state. Broad candidates must affect terminal total quality; targeted candidates must first affect gold/emitted source identity; transition candidates must first affect the next gold source, progress, or stop. Report necessity, partial sufficiency, rescue, side effects, and every matched control on complete sequences.

7 Why CoT Helps: The Causal Chain

The central result is not "an ablation lowers accuracy." It is the time-indexed sequence

$$I_{\text{route},k} \longrightarrow \pi_k \longrightarrow \Gamma_k \longrightarrow (\pi_{k+1}, \text{stop}) \longrightarrow Z \longrightarrow Y_{\text{full}}, \tag{6}$$

with side interventions $I_{\text{state},k} \rightarrow \Gamma_k$ and $I_{\text{terminal}} \rightarrow Z$. Here π_k is source-occurrence identity; coverage records misses, false positives, and repeated retrieval of the same occurrence; Γ_k includes the semantic progress state and any visible/KV ledger; and Y_{full} is the complete registered continuation, parsed count, exactness, and absolute error. A first-digit logit is not the primary endpoint.

Each intervention has a proximal claim gate. Corrupting a targeted-retrieval head at Q_k must first change the probability or identity of the gold next source before it is used to explain coverage or count error. Patching P_k must first change the next source, repetition, or stop before it is used to explain the final count. Patching Z may change Y_{full} without changing coverage and therefore isolates terminal readout. Clean-output rescue is evaluated beside corruption. Observational associations among coverage, state quality, and Y_{full} are descriptive; causal language is reserved for these paired activation interventions and their randomized controls.

Common-evidence readout control. Retrieval and readout must be separated without pretending that a teacher-forced list is a natural Direct computation. We therefore give both reasoning-policy conditions the same correct, passage-ordered evidence ledger and compare their state at the identical `Total:` token. This common-evidence intervention holds coverage fixed and is an auxiliary readout control: if the mode gap disappears, the natural advantage is consistent with retrieval mediation; if it remains, state construction or terminal readout contributes after evidence is fixed.

Compute and trace-content controls. An equal-length null scratchpad controls for extra tokens and recurrent computation. Counterfactual traces preserve token length and grammar while inserting a matched non-target, a repeat of an already visited source occurrence, or a false stop. A trace-length matched correct prefix controls for merely having more KV positions. These interventions test whether semantic trace content, rather than verbosity or index tokens, drives the benefit.

Noise with a measurable referent. In the main text, "noise" refers only to (i) measured retrieval/update errors in source selection, coverage, and stopping or (ii) conditional state dispersion in Equation (5). Attention entropy is a routing descriptor, not noise; parse, format, and truncation are behavioral failure categories. Larger N creates more opportunities for misses, duplicates, and stopping errors, but the shape of that accumulation is estimated rather than imposed by an independent-step product model.

TODO M1. On locked Qwen3-8B families, run a factorial route/state/terminal study with clean, corrupt, rescue, and sham levels. For every intervention, report the proximal endpoint first: gold/emitted source identity for route; next source/repeat/stop for state; complete answer for terminal. Then report unique coverage, false positives, state quality, sequence likelihood, exactness, and MAE with family-cluster intervals and paired randomization tests. Do not replace this chain with a single accuracy drop or a cross-sectional mediation regression.

TODO M2. Repeat the decisive links in Gemma4-E4B and run the common-evidence readout, equal-length null scratchpad, matched non-target, repeated-source, and false-stop controls. Claim functional replication if the same semantic links hold even when layers and head identities differ; otherwise state the model-specific boundary.

8 Cross-Model and Synthetic Tests

8.1 Functional replication

Qwen3-8B and Gemma4-E4B are co-primary mechanism models because both support the required reasoning interfaces and internal access. We do not expect identical head indices or a one-to-one circuit map. Cross-model support means that the same registered variables and causal links in Equation (6) hold. If only Qwen supports the full chain, the main claim is restricted accordingly and Gemma is reported as a boundary case.

8.2 Synthetic training dynamics

The synthetic experiments serve two roles: validating the causal analysis in a controlled model and revealing the formation order of the same functional motifs. At dense checkpoints we track behavior, held-out count/progress decoding, valid versus invalid update consistency, state-patch restoration, and the B_h, T_h, and U_h role scores. Decodability may emerge before causal use; a routing motif may emerge before a stable progress state. Those temporal separations are more informative than another final-checkpoint heatmap.

The v10 report supplies the clearest intervention vocabulary. Of the candidate final settings, v20 is the preferred basis because it adds query-conditioned target sets, guarded corpus regions, and a balanced count evaluation; v15 remains the all-sequence-loss control. The existing v20 run, however, has one training seed and switches objective after step 1500. Any change near that point is therefore an objective intervention, not evidence of a spontaneous phase transition. We do not splice v15 and v20 checkpoints into one trajectory.

TODO S1. Freeze a single v20-based generator, architecture, tokenizer, checkpoint schedule, and loss mask. Run at least three training seeds plus an all-sequence-throughout control matched to v15 exposure and an explicit objective-switch control. Use $N \leq 10$ for direct LLM comparability and $N = 11$–30 only as extrapolation. Save dense checkpoints before and after behavioral onset and the registered objective switch. At each checkpoint compute the same decodable–dynamic–causal ladder and B_h, T_h, U_h scores, using disjoint role-selection and causal-report families. Keep this section in the main paper only if emergence order and causal effects are stable across seeds; otherwise report it as supporting appendix evidence.

9 Related Work

Controlled transformer studies show that counting can be implemented by different algorithms and is sensitive to architecture, positional encoding, and optimization (Golkar et al., 2024; Yehudai et al., 2024; Chang and Bisk, 2025; Behrens et al., 2025). Studies of pretrained models report decodable count states, explicit partition-and-aggregate strategies, and possible geometric readout bottlenecks (Hasani et al., 2026a,b; Garcia, 2026). These results motivate our evidence hierarchy: count information can be present without being the state the model causally uses.

CoT and scratchpads can externalize intermediate computation (Nye et al., 2021; Wei et al., 2022). Mechanistic studies identify reusable attention motifs such as induction and successor heads (Olsson et al., 2022; Gould et al., 2023), while activation patching is sensitive to the corruption, metric, and patch location (Zhang and Nanda, 2024). Our focus is the same sparse semantic context under direct and explicit-reasoning interfaces, with source identity and unique coverage as registered mediators.

10 Limitations

Native reasoning traces are policy-dependent and need not faithfully expose all internal computation. Forced-prefix and activation interventions can be off-manifold; magnitude sweeps, natural donors, and side-effect audits are therefore required. Predicate availability and response-policy placement change what local states can identify and may also change the algorithm itself. The current behavioral summary also combines formal-v2 Direct/Native rows with v2.1 enumeration replacements and contains one sampled continuation per stimulus; source version and decoding variance must therefore be audited rather than hidden by pooling. The semantic roles may be shared across Qwen and Gemma even when their implementations differ, but two checkpoints cannot establish universality. Finally, synthetic emergence is supportive only when its task and metrics align with the LLM study.

11 Conclusion

The proposed paper is a causal comparison of counting computations, not a catalog of probes and attention maps. Direct answering is tested for a record-wise running update versus a terminal multi-record total; CoT is tested for matched-source retrieval and a semantic progress carrier. The strongest conclusion requires interventions to propagate through source identity, unique coverage, state quality, and the complete count. If only part of that chain is supported, the claim will be narrowed to that part rather than treating a decodable direction as a mechanism.

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A Claim Gates

The manuscript follows a claim ladder:

1. **Behavior:** trace-generating interfaces outperform Direct under matched stimuli.
2. **Process:** trace generation changes distinct-source retrieval, and the Direct total is identified as a running or terminal computation.
3. **Representation:** registered states satisfy decodable and dynamic tests.
4. **Component causality:** routing/update candidates selectively affect their semantic endpoints.
5. **Integrated causality:** those changes propagate through coverage and state quality to the complete count.
6. **Scope:** the functional chain replicates across Qwen and Gemma and emerges under a comparable synthetic setting.

Failure at a level narrows all claims above it. In particular, a decodable Direct record state without propagated patch effects is "available ordinal information," not a running counter; a CoT state that fails label controls is numeric-token scaffolding, not semantic progress.

B Prompt Serialization and Activation Manifest

The frozen V2 user message is serialized in the following order:

```
You will need to count all city-score
audit records in the passage below.
A city-score audit record names one
city and gives that city's numeric
score.
<passage>
[byte-identical frozen passage]
</passage>
How many city-score audit records are
in the passage?
[mode-specific response-policy block]
```

The response-policy block is the registered Direct, bullet enumeration, indexed enumeration, or Native-CoT instruction summarized in Table 1. Mechanism contrasts save both the raw user content and the output of the model-specific chat template; the latter, rather than the unrendered string, determines prefix identity.

Every activation row stores `family_id`, model and tokenizer revisions, mode, predicate/policy placement, raw-message and rendered-prefix hashes, complete input token IDs, source character spans, model-specific token spans, registered landmark indices, data seed, decoding seed, and `source_version`. Assertions verify byte-identical passages across modes, identical rendered token prefixes wherever a shared-prefix claim is made, equal sequence length for each clean/corrupt pair, and no truncation before any registered endpoint.

C Predicate and Policy Placement Matrix

Condition	Passage-time information	Valid use
Predicate-before, suffix-after (frozen V2)	Predicate known; visible suffix unseen; reasoning policy also unseen only if the rendered-prefix audit passes	Shared H_i candidate when prefix identity passes; otherwise analyze within mode; mode-specific Z_D, P_k
Predicate-after	Task predicate unseen	Content/position control at H_i; terminal tests remain valid
Predicate-repeat	Predicate known and refreshed after passage	Separates passage-time availability from recency
Policy-before (mechanism only)	Predicate and response policy known	Tests mode-specific passage encoding

Table 4: The causal mask and prompt serialization determine which state comparisons are identifiable. Conditions other than the first are mechanism controls, not extra cells in the frozen behavioral grid.

D Auxiliary Density-Code Hypothesis

One falsifiable implementation of H_{agg} is a normalized relevance statistic

$$\rho(N, L) = \frac{N}{N + \alpha(L - N)} + \eta.$$

Its inverse has sensitivity

$$N(\rho, L) = \frac{\alpha L \rho}{1 - (1 - \alpha)\rho}, \qquad \frac{\partial N}{\partial \rho} = \frac{\alpha L}{[1 - (1 - \alpha)\rho]^2}.$$

This model predicts that modest error in a density-like code can be amplified by context length. It is an auxiliary hypothesis, not the default description of Direct.

TODO A1. Compare density, unnormalized magnitude, and running-update models by held-out likelihood and by source-add/delete interventions. Retain this model only if its predicted changes in terminal state and absolute error are observed.

E Full Behavioral and Failure Tables

TODO A2. Include every attempted checkpoint, not only successful ones. For each checkpoint/mode report intent-to-treat exact accuracy, conditional numerical accuracy, parse failure, truncation, protocol failure, answer leakage, and invalid reasoning-mode rates. Record the precise reason any cell is excluded from mechanism analysis.

F Planned Result Figures

1. **Figure 1:** competing mechanisms, registered hidden states, operational head roles, and the time-indexed causal bridge (Figure 1).
2. **Figure 2 (TODO B1):** exact-accuracy and MAE surfaces plus failure budgets.
3. **Figure 3 (TODO D1–D2, R1–R3):** registered-state geometry, dynamic updates, and natural state-patch effects on the controlled sibling corpus.
4. **Figure 4 (TODO H1–H2):** real attention maps beside held-out ablation and patch effects for the three functional roles.
5. **Figure 5 (TODO M1–M2):** intervention-to-coverage-to-state-to-answer effects, including the common-evidence readout control.
6. **Figure 6 (TODO S1):** synthetic emergence order with seed uncertainty.

G Reproducibility Checklist

The final artifact will freeze model revisions, tokenizer files, chat templates, generation parameters, separate data and decoding seeds, stimulus and rendered-prefix hashes, parser code, character-to-token span maps, landmark definitions, intervention hooks, checkpoint IDs, `source_version`, and all discovery/validation/test splits. Request-level outputs are retained rather than only aggregates. Hardware, precision, batching, memory optimizations, and library versions are reported. Synthetic checkpoints and the exact data generator are stored with provenance.

H Detailed TODO Ledger

ID	Experiment	Required output and controls	Claim enabled
B0	Configuration and failure audit	Frozen revisions/templates/budgets/parsers; request-level parse, truncation, protocol, and leakage flags	Comparable behavioral panel
B1	Paired behavior grid	Four modes on 500 frozen stimuli and ten stimulus seeds; full requested model panel; exact count, registered success, full error budget, provenance, and family-clustered intervals	Robust behavioral phenomenon
B2	Held-out response surfaces	Qwen within-family primary; grouped held-out scoring; coefficient stability	Optional empirical law
D1	Controlled mechanism corpus	Tokenizer-specific equal-length siblings; stratified slots; family-disjoint splits; audited character/token spans and invariant delimiters	Identifiable representation and patching tests
D2	Prompt-policy controls	2×2 visible instruction by reasoning switch; predicate/policy placement; prefix hashes; paired decoding seeds and equal-length null scratchpad	Controlled algorithmic contrast
R1	Registered-state decoding	Grouped held-out probes; confound controls; state dispersion; correct/error split	Count/progress representation
R2	Direct dynamics and patches	Predicate-before $+1/0$ test; prefix/total shifts; single same-position donors; predicate-after negative control; terminal interventions	H_{run} versus H_{agg}
R3	CoT anti-leakage and state control	No-index/random/permuted labels; residual versus trace/KV transport; steering and natural patches	Semantic progress carrier
H1	Frozen role screen	B_h, T_h, D_h, U_h discovery/test split; raw and conditional mass	Candidate components
H2	Functional component tests	Local ablation; clean patch; path patch; semantic endpoints; matched controls	Necessity and partial sufficiency
M1	Integrated Qwen causal chain	Route/state factorial intervention; coverage, progress, full-count endpoints	Mechanism-to-behavior link
M2	Controls and Gemma replication	Common evidence; null scratchpad; semantic corruptions; co-primary replication	CoT explanation and scope
S1	Synthetic emergence	Frozen v20 basis; dense checkpoints; three seeds; all-sequence and objective-switch controls	Developmental support

Table 5: Unified TODO ledger. B0–B1, D1–D2, R1–R3, H1–H2, and M1 are required for the Qwen mechanism claim; M2 is additionally required for the stated Qwen–Gemma functional replication claim. B2 and S1 determine optional law and developmental scope.